# Race, explain and transfer

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Evaluate a model against measured performance and defend design decisions.

**SUCCESS CRITERIA - I can...**

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “relative deviation” and “energy budget” to explain your model or result. Sentence starters are optional.

**relative deviation**: difference from prediction compared with its magnitude

**energy budget**: an account of all energy stored and transferred

**revision**: a change made after checking evidence or reasoning

## 01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 6 times as large.
**Compare** ideal travel speed and state one real-world limitation.

**Word help:** circumference = distance around the edge of a circle

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

**Worked reasoning:** A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

**Your turn: Draw** an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

**Draw** or annotate your final car.
**Label** three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

**Marzano check - circle:** ① ② ③

## 04 Build the solution

A supplied car travels 5 m in 10 s.
**Find** measured speed. Its prediction was .60 m/s.
**Use** the supplied difference .10 m/s to calculate relative deviation=.10÷.60×100%.

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

**Use** your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A device receives 60 J mechanical energy at its shaft and produces electrical energy at its terminals.
**Identify** it and name one possible loss.

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

**Choose** one problem on this sheet.
**Circle** your current 1, 2 or 3 self-check and name the next step that would improve your explanation.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Propose one redesign and predict two possible effects, including a tradeoff.
**State** the measurement that would support or challenge your prediction.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3